
Energy References and Dissipation in Compactly Switched Tungsten Potentials

Computational study

Reproducible code, inputs, trajectories, and analysis supplied with this report

Abstract

Blending ground-state and electronically excited interatomic forces requires a rule for their relative weight. We define balanced-error compact switching (BECS): a monotone, continuously differentiable interpolation whose intermediate anchor minimizes the larger of normalized energy and force squared errors. Evaluations of two fixed tungsten ACE potentials on all 21,356 converged configurations in three distributed DFT archives reveal a critical calibration issue. The raw DFT energies and potential energies have different reference origins. On 487 intermediate-temperature validation configurations, the registered, unshifted criterion selects zero intermediate hot-state weight, giving energy and force RMSEs of 6.08319 eV/atom and 0.03800 eV/Å; it does not improve both errors over a declared sigmoid surrogate. Endpoint-based reference adjustments change the selected anchor substantially. Independently, velocity-reversal experiments show that force blending can either add or remove ionic energy at identical instantaneous excitation. The temperature driver's sharp neighbor cutoff also produces finite force jumps despite a smooth switch. Reduced-cell LAMMPS trajectories, analyzed with OVITO, expose large kinetic-energy growth and loss of a localized thermal-spike interpretation. The results establish neither a larger stable recoil range nor preserved damage suppression at the original simulation scale. They identify the reference conventions, driver regularity, and spatial containment needed before a calibrated switch can support those claims.

1 Motivation

Electronic excitation changes both the forces acting on ions and the channels through which deposited energy leaves an irradiated metal. These two effects need not be represented by the same approximation. A temperature-dependent interatomic potential changes the force field; an electron bath additionally carries energy, conducts heat, and exchanges energy with the lattice. Confusing these roles makes a smooth interpolation appear more physically complete than its equations justify.

Two-state layered potentials offer an economical representation of the first effect. The construction of [Wood et al. \(2026\)](#) combines fixed ACE potentials for tungsten at electronic smearing temperatures of 0.001 and 0.5 eV, with a local ionic temperature controlling the force weights. Their sigmoid is physically motivated, but its activation scale is parametric. A natural next step is to choose the interpolation by joint agreement with energies and forces at an intermediate electronic temperature. This paper makes that selection explicit and follows the resulting model through static calibration, implementation checks, and real MD diagnostics.

Our hypothesis was that balancing the two error channels and reducing unnecessary switch curvature would provide a more defensible activation law. The evidence supports a narrower conclusion. The registered calibration is dominated by an undocumented numerical energy-reference shift, while smooth force interpolation still lacks a sign-definite energy sink. The experiments therefore distinguish a precise, reproducible selection rule from an empirically validated radiation-damage

model. Negative outcomes, implementation repairs, and sensitivity analyses are retained rather than incorporated into an undisclosed change of objective.

Relation to existing approaches. Sommerfeld corrections introduce electronic-entropy contributions into interatomic free energies (Ackland, 2012). Two-temperature MD instead couples atoms to an electronic energy field (Duffy and Rutherford, 2007); tungsten cascade calculations show that electron–phonon coupling affects both peak damage and its relaxation (Zarkadoula et al., 2015). Electron-temperature-dependent forces can also compete with melting and structural transformations in excited tungsten (Murphy et al., 2015). These results motivate explicit separation of force modification from dissipative energy exchange.

Adaptive-precision methods blend models to allocate computational effort (Immel et al., 2025a). Hamiltonian extensions explicitly address energy and momentum conservation (Immel et al., 2025b). BECS does not reproduce that contribution: it retains the force-layering algebra and the two fixed ACE models. Its proposed ingredient is a joint error selection rule attached to a compact temperature map, together with an audit of what that rule can identify. The cubic interpolation and minimax optimization are standard mathematical tools; no novelty is claimed for either in isolation.

2 A compact switch with an explicit selection rule

2.1 Fixed force model and local temperature

The atom-wise force is

$$\mathbf{F}_i = (1 - \lambda_i)\mathbf{F}_i^{\text{gs}} + \lambda_i\mathbf{F}_i^{\text{ht}}, \quad 0 \leq \lambda_i \leq 1. \quad (1)$$

At uniform fixed weight, the corresponding energy is $U_\lambda = (1 - \lambda)U_{\text{gs}} + \lambda U_{\text{ht}}$. Both endpoints include the same ZBL overlay, so it appears exactly once in the blend. The supplied ACE coefficients, basis, and cutoffs are unchanged and evaluated with PACE (Lysogorskiy et al., 2021). LAMMPS hybrid/scaled implements Eq. (1) with atom-style variables and no derivative-of-weight force term. Such substyles contribute zero to LAMMPS’s potential-energy tally (LAMMPS developers, 2025); that tally is consequently not used as a blended energy or as evidence of energy conservation. Static energies and forces are evaluated independently on both endpoints.

For equal-mass tungsten atoms, the local driver is

$$T_i = \frac{m}{3M_i} \sum_{j \in \mathcal{N}_i} |\mathbf{v}_j - \bar{\mathbf{v}}_i|^2, \quad \bar{\mathbf{v}}_i = M_i^{-1} \sum_{j \in \mathcal{N}_i} \mathbf{v}_j, \quad (2)$$

expressed in energy units with the LAMMPS metal-unit conversion. The spherical neighborhood includes the central atom, has radius 5.28 Å, and contains M_i atoms. This matches the published definition, including its $3M_i$ denominator. The implemented compute returns Kelvin, which we multiply by k_B . The distributed example instead uses a 5.488 Å cutoff and passes its temperature output directly to thresholds labeled in eV; we resolve these ambiguities in favor of the manuscript definition and documented units. This prevents claiming exact reproduction of an undocumented author executable.

2.2 Balanced-error compact switching

For configuration c with N_c atoms, define the uniform-blend residuals

$$e_c(q) = \frac{E_c^{\text{gs}} + q(E_c^{\text{ht}} - E_c^{\text{gs}}) - E_c^{\text{DFT}}}{N_c}, \quad (3)$$

$$\mathbf{f}_{ci}(q) = \mathbf{F}_{ci}^{\text{gs}} + q(\mathbf{F}_{ci}^{\text{ht}} - \mathbf{F}_{ci}^{\text{gs}}) - \mathbf{F}_{ci}^{\text{DFT}}. \quad (4)$$

The configuration-balanced losses are

$$M_E(q) = \frac{1}{C} \sum_c e_c(q)^2, \quad M_F(q) = \frac{1}{C} \sum_c \frac{1}{3N_c} \sum_i \|\mathbf{f}_{ci}(q)\|^2. \quad (5)$$

We register the following one-dimensional convex optimization:

$$q^* = \arg \min_{q \in [0,1]} \max \left\{ \frac{M_E(q)}{M_E(0)}, \frac{M_F(q)}{M_F(0)} \right\}. \quad (6)$$

The normalization fixes the relative importance of energies and forces without a tunable channel weight. It minimizes the worst relative squared error, not the two channels separately. No energy offset is fitted in this registered calculation. Degenerate ties select the smallest minimizer; the present data do not produce a tie. Pooled Cartesian-force RMSE is reported separately to reveal sensitivity to configuration size.

Let $S(u) = 3u^2 - 2u^3$ on $[0, 1]$, extended by constants outside that interval. The electronic-temperature interpolation is

$$L(T_e) = \begin{cases} q^* S((T_e - 0.001)/0.099), & T_e \leq 0.1 \text{ eV}, \\ q^* + (1 - q^*) S((T_e - 0.1)/0.4), & T_e > 0.1 \text{ eV}, \end{cases} \quad (7)$$

where temperatures inside the numerical arguments are measured in eV. This curve is monotone and C^1 , with exact cold and hot anchors and zero slope at each knot. On each interval it uniquely minimizes $\int (L''(T_e))^2 dT_e$ given its endpoint values and zero endpoint slopes: the Euler–Lagrange equation is $L^{(4)} = 0$, and the four boundary conditions yield the cubic S . It is not generally C^2 at the intermediate knot.

For MD we use the separate ionic map

$$u_i = \frac{(T_i - b)_+}{a}, \quad \hat{T}_{e,i} = 0.001 + 0.499 \frac{u_i^2}{1 + u_i^2}, \quad \lambda_i = L(\hat{T}_{e,i}), \quad b = k_B(1000 \text{ K}). \quad (8)$$

Squaring the positive part makes the map derivative-continuous at b . It is exactly cold below b , approaches the hot endpoint as $T_i \rightarrow \infty$, and introduces no discontinuous recoil-energy dependence. The estimate $\hat{T}_e$ is a modeling coordinate, not an independently measured electron temperature.

The scale a is selected by matching $\chi = \int \sum_i \lambda_i dt$ to the $T^* = 0.5 \text{ eV}$ sigmoid on a fixed ground-state pilot. The resulting $a = 0.349931 \text{ eV}$ gives $\chi = 34.257081 \text{ atom ps}$ for both maps on seven retained pilot frames, through 0.263103 ps . This is an offline match. A new self-consistent trajectory can visit different configurations and temperatures, so its measured χ must be checked again. The short pilot includes adaptive stepping and a further 0.1 ps ; it does not stand in for a complete cascade.

A separate cold-activation check evaluates 100 independent canonical Gaussian velocity draws at 1000 K on the equilibrated ground-state geometry, using actual LAMMPS run-zero force evaluations. Neither map produces $\lambda_i > 1/2$ in any draw. The largest weights are 4.404×10^{-5} for the sigmoid and 9.245×10^{-33} for BECS, the latter arising from floating-point clipping roundoff. The largest sampled local temperature is 0.141770 eV . This tests thermal velocity fluctuations at a fixed geometry; it is not a long-time equilibrium trajectory, and overlapping atomic neighborhoods are not independent samples.

3 Electronic-temperature calibration and its reference dependence

3.1 Data, evaluation, and comparison maps

We evaluate every converged JSON configuration in the distributed VASP archives: 9,467 at the nominal cold endpoint, 2,444 at 0.1 eV , and 9,445 at the nominal hot endpoint. All paths under `NotConverged` are excluded: 15, 78, and 258, respectively. No new DFT calculation or potential refitting is performed. The source calculations use PBE, PAW W_{sv}-GW, a 700 eV cutoff, a $0.0268 \text{ \AA}^{-1}$ mesh spacing, and additional bands for smearing (Wood et al., 2026). Cell, position, and force rotations are applied consistently before LAMMPS evaluation.

The intermediate archive is partitioned by a fixed SHA256 rule on configuration paths: 1,957 configurations select q^* and 487 validate it. This is a within-archive split, not a guarantee of independence between related configurations. A leave-category-out analysis and a paired 1,000-resample configuration bootstrap accompany it. Each endpoint is evaluated on its own training archive and is not an independent accuracy test. All 41 blend fractions $0, 0.025, \dots, 1$ are retained; their energies and forces are exact linear combinations of independently evaluated endpoints.

A metadata audit finds 34 cold-archive and 15 hot-archive entries labeled with smearing 0.1 eV . The intermediate archive itself is uniformly labeled 0.1 eV , and no exact joint energy/force/geometry record matches either endpoint archive. Nevertheless, the endpoint temperature labels are not uniform;

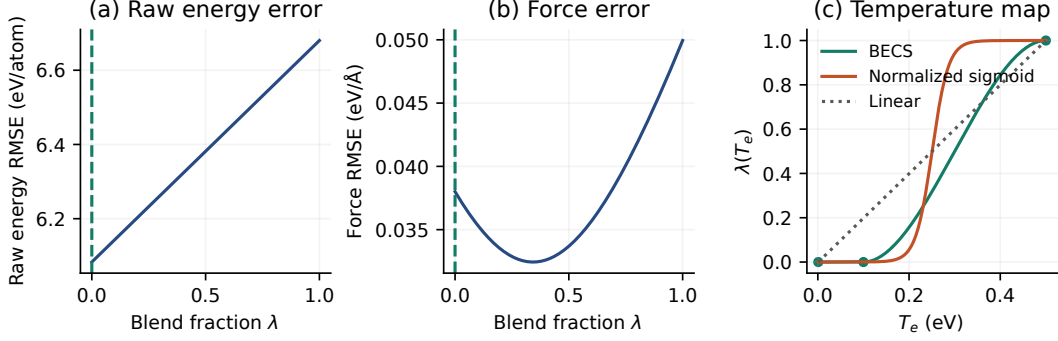


Figure 1: Complete intermediate-temperature calibration diagnostics. Panels (a,b) show validation losses across the full blend interval; the registered selection is $q^* = 0$. Panel (c) compares the resulting compact electronic map with explicitly defined comparison maps. The different energy and force trends are preserved, rather than selecting only a favorable error channel.

Mapping at $T_e = 0.1$ eV	λ	Energy (eV/atom)	Force (eV/Å)	Force (pooled) (eV/Å)
BECS (registered)	0.00000	6.08319	0.03800	0.07565
Normalized sigmoid	0.00021	6.08331	0.03799	0.07564
Linear electronic map	0.19840	6.20159	0.03349	0.06825
Quadratic electronic map	0.04000	6.10705	0.03685	0.07385
Identity sigmoid, $T^* = 0.5$	0.00001	6.08319	0.03800	0.07565
Hot endpoint	1.00000	6.68006	0.04994	0.08494

Table 1: Validation on all 487 reserved intermediate configurations. “Force” weights configurations equally; “pooled” weights Cartesian force components equally. Energies retain the raw JSON reference. The sigmoid extensions are static-coordinate diagnostics, not the published ionic baseline.

filtered endpoint diagnostics are supplied. This uncertainty is distinct from the calibration/validation split and prevents an unqualified claim about the original fitting membership of every structure.

The published sigmoid is a function of T_i with $T^* \neq T_e$. Static DFT geometries do not specify ionic velocities. Thus the paper does not define a unique sigmoid accuracy curve against T_e . We explicitly distinguish two diagnostic extensions: an identity map that inserts T_e into the ionic sigmoid, and an endpoint-normalized sigmoid

$$L_{\text{sig}}(T_e) = \frac{s(x) - s(0)}{s(1) - s(0)}, \quad x = \frac{T_e - 0.001}{0.499}, \quad s(x) = \frac{1}{1 + \exp[-28(x - 1/2)]}. \quad (9)$$

Equation (9) is a declared surrogate, not a reproduced physical electron-to-ion mapping. Linear and quadratic electronic maps supply additional transparent references. Actual MD baselines use the published ionic sigmoid and all six reported thresholds, with $\sigma = T^*/14$.

3.2 A failed raw-energy calibration

The registered energy and force quadratic coefficients on the calibration split are

$$M_E(q) = 0.35805161q^2 + 7.26623715q + 37.00376616, \quad (10)$$

$$M_F(q) = 0.00330903q^2 - 0.00266544q + 0.00174980. \quad (11)$$

Energy error increases throughout the admissible interval. Equation (6) therefore selects $q^* = 0$. This eliminates the lower branch of the electronic interpolation; it does not eliminate hot-state forces in MD, where the upper branch and the exposure-matched ionic map remain active. The separate force optimum is 0.40275 on the calibration split and 0.39069 on all intermediate configurations; pooling force components changes the latter to 0.52880. The reported reference minima, approximately 0.2 for energy and 0.35 for force, are from Supplemental Fig. 18 of Wood et al. (2026). They are not reproduced by the raw archive energy convention and the declared weighting here.

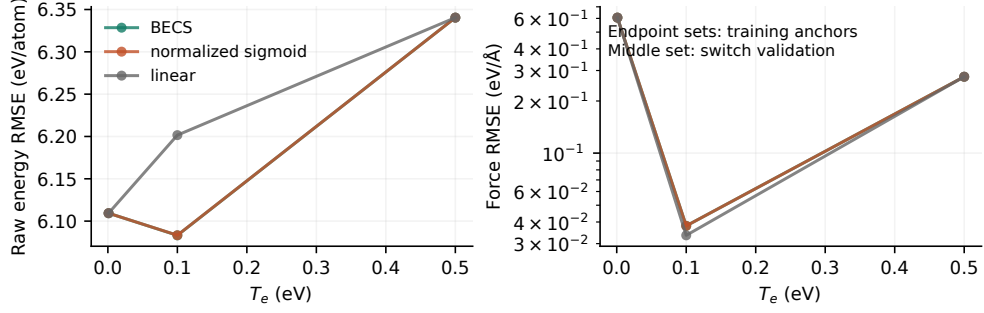


Figure 2: Accuracy at the three available electronic temperatures. Endpoint errors use their respective training archives; the intermediate point uses the switch-validation split. Different temperatures contain different configuration distributions, so connecting lines are visual guides, not evidence of smooth error versus temperature on matched geometries. The normalized sigmoid is defined in Eq. (9).

Model	Published E (meV/atom)	Centered E (meV/atom)	Published F (eV/Å)	Full archive F (eV/Å)	Listed groups F (eV/Å)
GS	47.50	42.23	0.638	0.63679	0.63759
HT	35.56	31.33	0.038	0.22722	0.04111

Table 2: Endpoint checks against Table I of Wood et al. (2026). Centered energies remove the full-archive mean residual as a post-evaluation reference diagnostic; the author’s energy shifts are unknown. Our force columns pool Cartesian components, over either the full archive or categories listed in source Table II. Category restriction is secondary and does not change the primary evaluation. Exact file membership and aggregate error conventions differ.

Table 1 makes the negative result explicit. Against the normalized sigmoid surrogate, BECS changes validation energy RMSE by only -0.000128 eV/atom and force RMSE by $+0.00000644$ eV/Å. The paired 95% bootstrap intervals are $[-0.000129, -0.000127]$ eV/atom and $[0.00000480, 0.00000803]$ eV/Å, respectively. The force change has the wrong sign for a joint improvement. These intervals describe resampling variability within this archive and do not cover reference-convention or structural-distribution uncertainty.

The raw energy RMSE cannot be interpreted as the potential’s physical formation-energy accuracy. The author response accompanying Wood et al. (2026) states that each electronic-temperature DFT set was shifted to match its minimum-energy structure to a cohesive energy. Those numerical shifts are not included with the supplied coefficients or JSON energies. For example, the cold endpoint has a mean energy residual of 6.109206 eV/atom but a centered residual standard deviation of 0.042234 eV/atom. The corresponding hot values are 6.340328 and 0.031334 eV/atom. A large common offset, rather than comparably large variation between configurations, dominates the raw energy loss.

The force comparison also needs qualification (Table 2). The cold pooled force RMSE closely matches the published value, whereas the full-archive hot value does not: 0.22722 versus 0.038 eV/Å. Excluding temperature-tag mismatches does not resolve this gap. Source Table II supplies group weights, but omits the two large gamma-surface categories from its listed groups and omits interstitials for the hot fit. Restricting our diagnostic to its listed categories gives 0.63759 and 0.04111 eV/Å for the cold and hot models, respectively. Thus category coverage explains much of the hot discrepancy. Several per-group counts still differ from the distributed archives, and the exact global aggregation and file membership remain unidentified. Numerical force verification establishes implementation consistency; it does not establish exact reproduction of the published hot-endpoint accuracy.

Energy reference	Selected q	Energy RMSE (eV/atom)	Force RMSE (eV/Å)
Raw, registered	0.0000	6.08319	0.03800
Common cold-anchor shift	0.0848	0.02692	0.03570
Separate endpoint shifts	0.1356	0.02605	0.03457

Table 3: Post-evaluation energy-reference sensitivity. Offsets use only endpoint archives. They do not refit the ACE coefficients or change forces, and are not substituted for the registered calibration or used in MD. Separate endpoint offsets also change the relative energy origin of the two surfaces. These are alternative conventions, not recovered author constants.

3.3 Why an unspecified energy origin changes the answer

Write $M_E(q) = Aq^2 + Bq + C$ and let $\bar{d}$ be the mean endpoint energy difference per atom. A common shift c of both model energies changes neither endpoint force nor the force difference, but gives

$$q_E(c) = -\frac{B + 2c\bar{d}}{2A} \quad (12)$$

for the unconstrained energy optimum. Whenever $\bar{d} \neq 0$, its location depends on the chosen energy origin. This is a problem of reference alignment, not a physical failure of adding a constant to a potential. A calibration using absolute residuals becomes interpretable only after the model and DFT energy references have been specified consistently.

We therefore retain the failed raw-energy selection and separately examine two explicit adjustments (Table 3): a common offset estimated from the cold endpoint, and distinct mean offsets estimated from the two endpoint archives. The selected anchor changes to 0.0848 and 0.1356. The apparent improvement under these adjusted conventions cannot establish an improvement under the author’s convention, which remains numerically unidentified. Neither adjustment was used to rescue the main result after observing validation errors.

There is also a simple obstruction to joint improvement in the raw convention. On the validation set, energy error is strictly increasing on $[0, 1]$, whereas force error decreases from zero up to its interior optimum. The surrogate’s small positive blend lies between these optima. Moving left improves energy and worsens force; moving right worsens energy. No alternative *uniform* scalar weight at this temperature can strictly improve both channels over that point. This statement is specific to the measured loss curves and reference convention. It does not rule out a better reference-aligned calibration or a richer, configuration-dependent model.

4 Dynamical tests and the limits of a small simulation cell

4.1 Protocol and explicit deviations

The radiation-damage calculation uses single-crystal W, an isotropic primary knock-on velocity, and identical recoil directions across methods. A ground-state and a Marinica EAM cell are equilibrated separately; all ACE variants start from the same ground-state configuration. Cube-face layers occupying the outer 2.5% of each dimension are thermostatted at 1000 K, with NVE integration in the interior during the cascade. Adaptive steps range from 10^{-9} to 5×10^{-4} ps, update every 100 steps, and limit displacement to 0.05 Å. After recovery of the upper timestep and one further adaptive block, the whole cell enters a 30 ps NVT anneal. The EAM reference uses the supplied Marinica potential (Marinica et al., 2013).

The initial production plan covered 1, 2, 5, 10, 15, 20 keV with 25 directions at each point. The full production study was not executed; a direct capacity audit below quantifies its cost. The actual diagnostics use 8^3 conventional BCC cells, or 1,024 atoms, at 1, 2, 5 keV with three paired directions per method. Their equilibration is 1 ps of NPT followed by 1 ps of NVT, rather than the published 100 ps. The equilibrated ACE cell edge is 25.478026 Å, and the EAM cell edge is 25.331374 Å. These are stress tests of the pipeline and switching dynamics, not a substitute for a spatially converged radiation-damage calculation.

An additional one-direction series at 1 keV checks the six published sigmoid thresholds $T^* = 0.5, 0.75, 1.25, 2.5, 3.75, 5.0$ eV. Every sigmoid retains $\sigma = T^*/14$. The $T^* = 0.5$ eV member repeats the same seed as the main diagnostic and is not counted as an independent repetition. MD uses PACE’s recursive contraction order; comparison against the product evaluator and a force finite-difference check establish numerical equivalence without altering the ACE basis or coefficients. Execution has a registered one-hour wall limit per trajectory; any stopped trajectory is censored, not counted as a completed anneal or a physical instability.

Structural measurements. OVITO polyhedral template matching identifies non-BCC atoms with RMSD cutoff 0.1. An alpha-shape surface mesh uses the published 10 Å probe radius. Because OVITO cannot tessellate that probe directly in the small diagnostic box, we periodically replicate the classified configuration $2 \times 2 \times 2$ for analysis and divide extensive volumes by eight. This preserves the periodic point set and radius while leaving the physical finite-size limitation unchanged. We report both the volume at the end of adaptive stepping and the maximum over retained frames. Samples are spaced by 100 steps during adaptive integration and 1,000 steps during annealing; these are sampled maxima, not guaranteed continuous-time maxima.

Final vacancies and interstitials are counted using OVITO Wigner–Seitz analysis against the corresponding ideal BCC reference, affinely mapped to its cell (OVITO developers, 2026). Both equilibrated starting cells have zero vacancies, zero interstitials, and no non-BCC atoms at the stated PTM cutoff. Equal final occupancy counts are reported as Frenkel pairs. A secondary translation audit estimates the crystal displacement from the phases of the (110), (101), and (011) density Fourier components and repeats Wigner–Seitz analysis after removing that uniform translation. Raw and registered counts are both retained, with no model-specific alignment parameter. This audit was introduced after the first EAM result revealed a large drift sensitivity, and is applied to every completed trajectory. Weak crystalline order would make either site-based interpretation questionable. Exposure is integrated from the actual variable timestep and logged η . The operational volume-growth statistic is the least-squares slope over post-adaptive volume samples. Slopes below 10^{-10} Å³/ps are treated as numerical zero; without this tolerance, roundoff in a constant cell-filling volume changes its apparent sign. The publication stops a series when more than 25% of 25 trials have increasing volumes. Our smaller diagnostic cohort and percolating spikes do not identify that production threshold.

4.2 Measured outcomes and comparison with the published regime

The diagnostic sweep contains 36 trials: 31 complete the anneal, 5 terminate with a lost-atom error, and 0 remain wall-censored. Table 4 reports measurements from completed anneals; Table 5 reports every lost-atom trajectory separately. The 5 keV sigmoid cohort has no completed trajectory and therefore no final defect statistic. The 5 keV BECS entry is its one completed trajectory, with raw and registered occupancies of 237 and 239; it is not a three-direction estimate. Its final kinetic temperature is 2648 K and its low crystal-density amplitudes make a point-defect interpretation inappropriate. Individual measurements and last recorded times are retained in `md_all_trials.csv`. The six threshold pilots are listed separately in Table 6.

Offline exposure matching remains approximate in self-consistent dynamics. Ratios of mean BECS to sigmoid exposure for completed paired cohorts are 1 keV: 1.008; 2 keV: 1.012. Exposure is close on these cohorts, without any further adjustment of the activation scale using their outcomes. BECS has nonzero excitation in actual recoil trajectories despite its zero intermediate anchor; its dynamical failure is not hidden by replacing it with ground-state-only forces.

The surface analysis reaches full periodic-cell filling in 34 of the 36 main trials. The volume curves consequently do not establish reduced peak thermal-spike volume. In this geometry, loss of confinement invalidates a literal extrapolation of the published volume-growth rule. A completed anneal is an execution outcome, not a bulk stability certificate. Lost atoms constitute a separate numerical failure and are not silently recoded as the published volume criterion.

Figure 4 separates transient recoil excitation from sustained kinetic-energy growth. The largest recorded whole-cell kinetic temperature across the diagnostic sweep is 1.51×10^7 K. This thermodynamic output includes any collective motion, whereas the local switching driver removes each neighborhood’s COM velocity. It is an integration diagnostic, not an equilibrium material temperature or a predicted experimentally realizable state. The exposure accumulated more than 1 ps after the

E (keV)	Model	Complete/lost (of 3)	Raw WS (pairs)	Registered WS (pairs)	Peak volume (nm ³)	Complete χ (atom ps)
1	GS	3/0	63.7 $\pm$ 92.3	2.0 $\pm$ 1.0	16.54 $\pm$ 0.00	0.0 $\pm$ 0.0
1	EAM	3/0	43.0 $\pm$ 35.5	2.7 $\pm$ 1.2	16.25 $\pm$ 0.00	0.0 $\pm$ 0.0
1	Sigmoid	3/0	102.7 $\pm$ 87.3	1.7 $\pm$ 0.6	15.76 $\pm$ 1.35	40.3 $\pm$ 5.6
1	BECS	3/0	118.3 $\pm$ 119.3	1.3 $\pm$ 1.2	16.19 $\pm$ 0.61	40.6 $\pm$ 5.6
2	GS	3/0	4.0 $\pm$ 2.6	4.0 $\pm$ 2.6	16.54 $\pm$ 0.00	0.0 $\pm$ 0.0
2	EAM	3/0	3.0 $\pm$ 1.0	3.0 $\pm$ 1.0	16.25 $\pm$ 0.00	0.0 $\pm$ 0.0
2	Sigmoid	3/0	18.3 $\pm$ 25.7	4.0 $\pm$ 1.7	16.54 $\pm$ 0.00	87.9 $\pm$ 14.7
2	BECS	3/0	4.3 $\pm$ 5.8	1.7 $\pm$ 1.2	16.54 $\pm$ 0.00	88.9 $\pm$ 13.7
5	GS	3/0	7.0 $\pm$ 2.0	7.3 $\pm$ 2.5	16.54 $\pm$ 0.00	0.0 $\pm$ 0.0
5	EAM	3/0	5.0 $\pm$ 2.6	5.0 $\pm$ 2.6	16.25 $\pm$ 0.00	0.0 $\pm$ 0.0
5	BECS	1/2	237.0	239.0	16.54	1441.1

Table 4: Completed finite-cell anneals only. Values with spreads are means and sample standard deviations of three directions. The 5 keV BECS row contains one survivor, so no spread is estimated; its occupancies are not interpreted as crystalline point defects. Complete/lost counts refer to all three attempted directions. Every exposure and peak in this table comes from a completed history. The failed 5 keV sigmoid cohort is reported in Table 5.

Model	Direction	Last sample (ps)	Observed χ (atom ps)	Sampled peak (nm ³)	Last T (K)
BECS	0	6.631	1659.54	16.54	1417
BECS	1	26.218	422.66	16.54	7118297
Sigmoid	0	5.481	1411.91	16.54	2084
Sigmoid	1	2.518	928.18	16.54	13732
Sigmoid	2	29.317	1872.52	16.54	3250045

Table 5: All five lost-atom trajectories, each at 5 keV. Time is the last retained thermo sample. Exposure and peak volume cover the observed history only and are not full-anneal values. No final Frenkel-pair count exists after loss of atoms. These are terminal failure records, not unfinished continuations.

adaptive phase is recorded separately for every trajectory, so a brief initial spike is not conflated with a long excited-state residence.

Structural reference alignment is also consequential. The first 1 keV EAM trial has 78 raw vacancy/interstitial pairs and 2 after uniform-translation registration; the corresponding first ground-state trial changes from 4 to 3. The unregistered ground-state count in another 1 keV direction changes from 170 to 2. The registered crystal-density amplitudes are retained to distinguish a rigid shift from loss of crystalline order. Momentum deposited into a small periodic cell can translate the whole crystal; Eq. (1) can add further net force when weights vary spatially. Neither effect should be mistaken for point-defect production.

For 19 executions the one-hour cap was followed by an explicitly recorded continuation from the last saved positions and velocities at text-dump precision. The thermostat chain was reinitialized, and the superseded partial tail was excluded from analysis. Of these, 17 reach the cumulative 30 ps anneal; failures remain in the outcome counts. The continued runs are not bitwise uninterrupted trajectories, and these additional protocol deviations are retained in each run record.

Measured production cost. To audit the unexecuted sweep directly, we ran the unchanged models in 250,000- and 2,000,000-atom cells on 32 MPI ranks, with 20 measured integration steps per model and cell size. All eight benchmarks completed. Their maximum aggregate process RSS was 30.18 GiB; memory was not the limiting resource. Table 7 gives measured step times and projected anneal costs. The four-series sweep (GS, EAM, sigmoid 0.5 eV, BECS), with 25 directions at each of 1, 2, 5, 10, 15, 20 keV, projects to 421 node-days for annealing alone. Including all six sigmoid thresholds projects to 1231 node-days. These estimates use the smaller box through 10 keV and the larger box at 15 and 20 keV; they exclude equilibration, adaptive integration, output, and analysis. They are engineering projections from short integration runs, not measured cascade runtimes. The

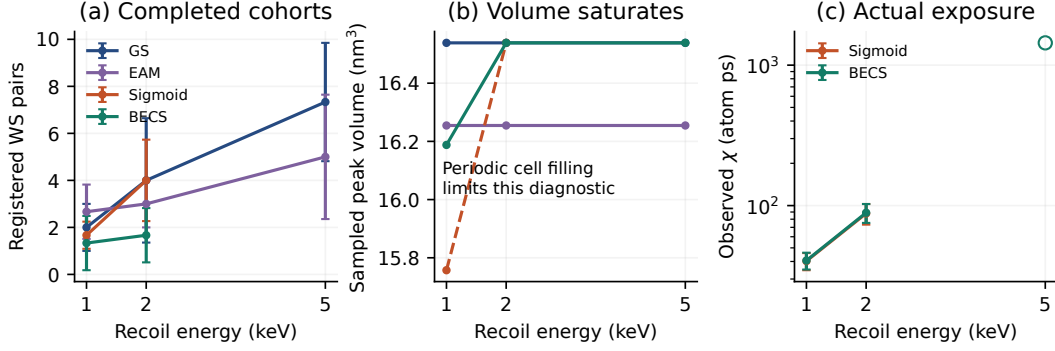


Figure 3: Real reduced-cell simulations. (a) Registered defect counts for fully completed three-direction cohorts, with sample standard deviations. Raw counts and the single disordered survivor remain in Table 4. (b) Completed-history peaks reach the periodic cell volume, invalidating an isolated-spike interpretation. (c) Complete-history exposure; the open circle is the single 5 keV BECS survivor, without a cohort spread. Failed histories are confined to Table 5.

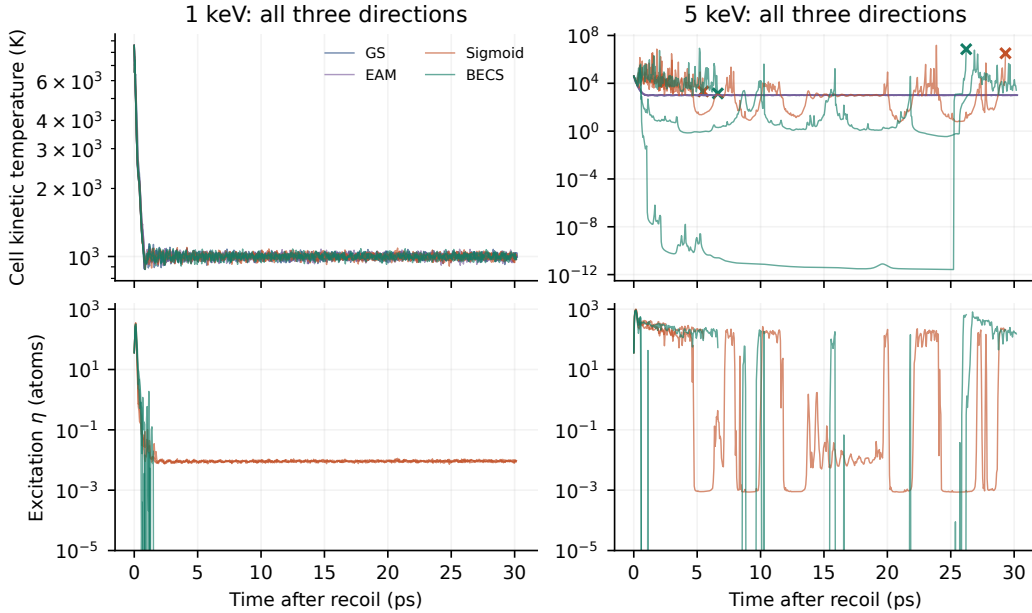


Figure 4: Recorded temperature and excitation histories at the lowest and highest diagnostic energies. Each thin curve is one of the three directions; crosses mark lost-atom termination. All directions are shown, including failures. Excitation below the displayed range is omitted from view but retained in the logs. The common color identifies a force model, not an average trajectory. Large kinetic energies and sustained excitation are observed together, without identifying a unique causal contribution from force blending, finite size, or thermostat response.

full sweep was not launched on this CPU allocation. No unexecuted energy is assigned a defect count, exposure, or stability outcome.

4.3 Numerical comparison with the published reference

Figures 5, 6, and 7 of Wood et al. (2026) contain raster plots rather than distributed numerical arrays. We extract the embedded images from the supplied PDF, calibrate their logarithmic and linear axes against tick locations, and retain every resolvable colored marker. Pixel coordinates, color masks, axis fits, and visual overlays are supplied. An uncertainty envelope of four pixels is used for exposure and volume; eight pixels is used for partly obscured defect markers. These envelopes quantify

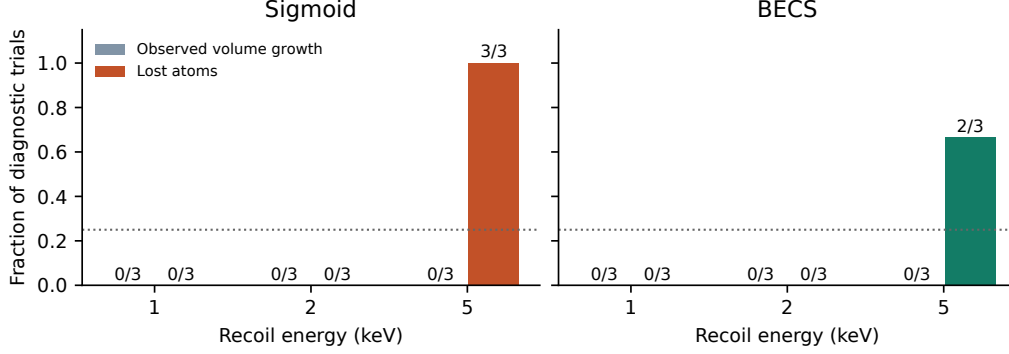


Figure 5: Why the diagnostic trajectories cannot determine a stable recoil range. Recorded post-adaptive volume growth detects no events, even when atom loss terminates most or all 5 keV runs. The dotted line is the published 25% criterion, whose intended ensemble has 25 directions. Cell-filling volumes have saturated and failed histories are truncated. Thus neither zero observed growth nor completed anneals certify bulk stability.

T^* (eV)	Raw WS	Registered WS	Peak (nm ³)	χ (atom ps)	Outcome
0.5	5	1	16.54	34.967	complete
0.75	1	1	16.54	19.293	complete
1.25	1	1	16.54	10.337	complete
2.5	2	2	16.54	4.915	complete
3.75	1	1	16.54	3.148	complete
5	127	1	16.54	2.303	complete

Table 6: One-direction 1 keV implementation pilots at every published T^* , always with $\sigma = T^*/14$. These are actual small-cell runs. The 0.5 eV case duplicates a main-sweep seed and is not an additional independent trial. This table cannot estimate a production stability range.

reading precision, not ensemble variation or statistical confidence. Fully occluded markers cannot be recovered and are not invented. Figure 6 presents these source observations separately from our simulations.

The exposure comparison reveals an additional reproducibility issue. For nonnegative kinetic temperature and the stated $\sigma = T^*/14$,

$$\lambda_{\text{sig}}(T_i) \geq (1 + e^{14})^{-1}, \quad \chi_{\text{anneal}} \geq \frac{N \cdot 30 \text{ ps}}{1 + e^{14}}. \quad (13)$$

At $N = 250,000$, this lower bound is 6.236460 atom ps, even at zero kinetic temperature; the cascade can only add exposure. Every digitized 1 keV source value lies below it, outside the reading envelope. Consequently the plotted source axis cannot be equated to an unnormalized, all-atom, full-30-ps integral in atom ps without an additional convention. A unit conversion, subtraction, normalization, or different integration interval could resolve the discrepancy, but no such choice is imposed here. Figure 6 also labels exposure at 30 ps while the accompanying prose describes the adaptive-end integral. These ambiguities prevent a numerical claim of matched aggressiveness against the published curves; the measured self-consistent matches in our diagnostic cohorts remain comparisons against our own sigmoid implementation.

The source text states a highest recoil of 15 keV at $T^* = 0.5$ eV, whereas its Fig. 5 contains a red marker near 20 keV. This distinction could separate the last accepted energy from a termination trial, but per-trial source outcomes were not supplied. For this reason neither marker extent nor textual reach is converted into a reconstructed seven-of-25 stability decision. The 200 keV source response gives 1203.4 pairs for $T^* = 5$ eV and 1062.76 for GS; that documented high-energy exception remains part of the reference.

Model	250,000 atoms		2,000,000 atoms	
	s/step	h/30 ps	s/step	h/30 ps
GS	0.1917	3.19	1.5356	25.59
EAM	0.0056	0.09	0.0446	0.74
Sigmoid	0.4645	7.74	3.7382	62.30
BECS	0.5487	9.15	4.3830	73.05

Table 7: Actual production-size throughput measurements on 32 MPI ranks. Seconds per step are measured; hours per 30 ps are projections for 60,000 steps. The sigmoid benchmark uses $T^* = 0.5$ eV and the BECS benchmark retains the frozen calibration and activation scale. These runs test capacity, not radiation damage.

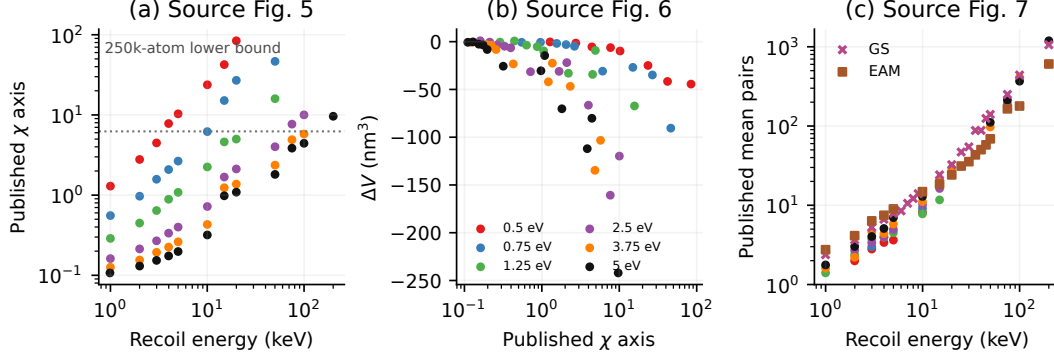


Figure 6: Numerical digitization of the supplied paper, not new cascade simulations. All resolved source markers are retained; overlap hides some values in panel (c). Panel (b) is the published change in peak volume relative to GS, not the absolute peak. The horizontal line in (a) is the literal full-anneal sigmoid exposure bound for the smallest published atom count. Pixel uncertainty is tabulated for the 1 keV comparison in Table 8 and stored for every point.

5 Mechanism: smooth force mixing is not a damping law

Writing $\Delta \mathbf{F}_i = \mathbf{F}_i^{\text{ht}} - \mathbf{F}_i^{\text{gs}}$, define instantaneous power relative to the ground-state force field,

$$P_{\text{mix}} = \sum_i \lambda_i (\mathbf{F}_i^{\text{ht}} - \mathbf{F}_i^{\text{gs}}) \cdot \mathbf{v}_i. \quad (14)$$

This quantity is evaluated from both endpoint forces on the same saved configuration, not inferred from LAMMPS's atom-scaled energy tally.

Proposition 1. *For a switch depending only on COM-corrected local kinetic temperatures, nonzero switching power cannot be nonpositive for every phase-space state.*

Proof. Replacing every velocity by its negative reverses each neighborhood COM velocity and leaves all squared relative velocities unchanged. Thus T_i , λ_i , and the endpoint force differences are unchanged, while Eq. (14) changes sign. Any state with nonzero power has a partner with positive switching power. $\square$

This elementary symmetry is not a new conservation formalism. It isolates a limitation of interpreting temperature-dependent force interpolation as universal drag. A friction force proportional to $-\mathbf{v}$ changes sign with velocity; its dissipative power does not. Two-temperature energy exchange and force-field interpolation serve different purposes.

On this configuration, the sigmoid gives $P_{\text{mix}} = 697.822$ eV/ps and BECS gives 679.938 eV/ps; reversal gives the corresponding negative values. A different ground-state pilot snapshot gives negative powers for both maps. Across ten map/snapshot combinations, direct LAMMPS forces agree with Eq. (1) within 3.12×10^{-12} eV/Å. Implemented weights agree with the Python definition within 8.33×10^{-16} , and a uniform velocity boost changes weights by at most 3.56×10^{-15} . The reversal residual in power is below 1.32×10^{-11} eV/ps. These checks distinguish physical nonconservation from an incorrect force implementation.

T^* (eV)	Published, digitized		Reproduced, one direction	
	Plotted χ	ΔV (nm ³)	χ (atom ps)	ΔV (nm ³)
0.5	1.295 ± 0.061	-0.2 ± 1.6	34.967	0.000
0.75	0.556 ± 0.026	-0.6 ± 1.6	19.293	0.000
1.25	0.289 ± 0.014	-0.4 ± 1.6	10.337	0.000
2.5	0.162 ± 0.008	0.9 ± 1.6	4.915	0.000
3.75	0.126 ± 0.006	-0.6 ± 1.6	3.148	0.000
5	0.107 ± 0.005	-0.6 ± 1.6	2.303	0.000

Table 8: Both references at 1 keV and all six sigmoid thresholds. Published values are read from Figs. 5 and 6; the spreads shown are digitization envelopes. The reproduced values come from the six completed 1,024-atom implementation pilots, each paired with the same GS direction for ΔV . Their cell-filling peaks make these volume differences uninformative about bulk damage. Published and reproduced ensembles and cell sizes differ. No statistical error bar can be estimated from one reproduced direction.

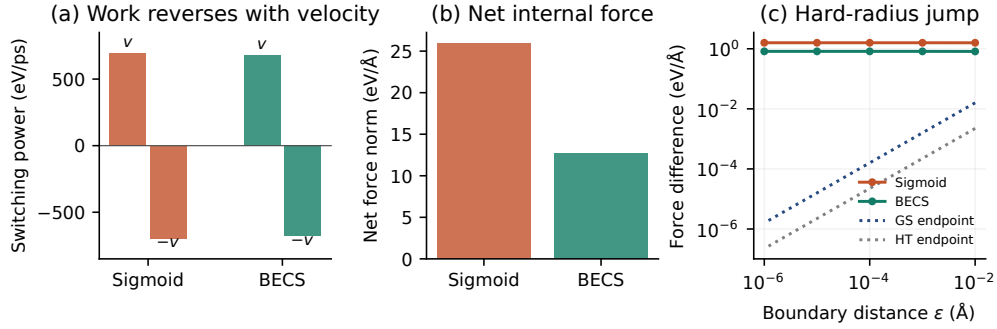


Figure 7: Force reevaluation from a saved 1 keV sigmoid configuration at the end of adaptive stepping. (a) Velocity reversal changes the sign of switching power without changing weights. (b) Spatial force scaling produces a net internal force. (c) A selected neighbor is placed just inside and outside the temperature-neighborhood radius. Endpoint force differences vanish with displacement, but blended force differences approach finite limits. These are snapshot and continuity probes, not integrated cascade energy balances.

Momentum cancellation provides another diagnostic. Although each endpoint has zero net internal force up to roundoff, $\sum_i \lambda_i \Delta \mathbf{F}_i$ need not vanish for spatially varying weights. On the configuration in Fig. 7, net-force norms are 25.9483 eV/Å for the sigmoid and 12.7485 eV/Å for BECS. Reducing this norm on a snapshot does not prove smaller long-time drift. A corresponding energy-coupled Hamiltonian requires additional terms and constitutes a different model (Immel et al., 2025b).

Smooth weights do not guarantee continuous forces. The fixed local-temperature compute includes a neighbor if its distance is strictly below $r_c = 5.28$ Å. Both the neighborhood count and velocity variance can therefore jump when an atom crosses this boundary. Continuous endpoint forces then imply

$$\lim_{\epsilon \downarrow 0} [\mathbf{F}_i(r_c + \epsilon) - \mathbf{F}_i(r_c - \epsilon)] = [\lambda(T_i^{\text{out}}) - \lambda(T_i^{\text{in}})] \Delta \mathbf{F}_i(r_c). \quad (15)$$

Derivative continuity of L does not remove a finite jump in its input. A secondary constructive test holds velocities fixed and moves one neighbor across the radius in the same saved configuration. The pair is selected, before force reevaluation, for the largest predicted inclusion/exclusion weight change among pairs within 0.005 Å of the boundary. It is an existence example, not a frequency or typical-impact estimate.

With $\epsilon = 10^{-6}$ Å, the measured local-temperature jump is -0.244746 eV. The sigmoid and BECS weight jumps are -0.841206 and -0.434985 ; their force-difference norms are 1.58668 and 0.82047 eV/Å. In contrast, the corresponding ground-state and hot-endpoint force changes are 1.59×10^{-6} and 2.23×10^{-7} eV/Å. The full perturbation sequence $\epsilon = 10^{-2}, \dots, 10^{-6}$ Å confirms the finite limiting behavior in Fig. 7c. This diagnoses another reason that smoothing the scalar switch alone

cannot guarantee stable MD. It does not identify how much these crossings contribute to the observed heating, and the prescribed temperature driver is not modified in any trajectory.

Finally, χ measures exposure, not dissipated energy. Equation (14) contains force–velocity correlations absent from $\int \sum_i \lambda_i dt$. Two trajectories with identical χ can therefore perform different work. The velocity-reversal test makes this failure of sufficiency explicit without assuming a particular instability mechanism. Long hot-state residency can contribute to feedback in a cascade, but neither residency alone nor a reduced switch slope guarantees stable integration or physical stopping.

6 Failure analysis and implications

Four limitations restrict interpretation. First, the primary energy calibration uses incompatible reference origins. The author-specified cohesive-energy referencing is a substantive part of the model definition, not a cosmetic plotting offset. Reporting smaller force error after choosing a different energy convention would not validate the registered joint criterion. The separate reference-sensitivity calculations therefore remain diagnostic. In particular, their improvement cannot be promoted to a result for the MD candidate, which retains the registered anchor.

Second, structural coverage and weighting matter. Of the 2,444 intermediate configurations, 1,943 belong to the `slice_sample` category, whose ground-state force RMSE is 1.41×10^{-15} eV/Å. The intermediate set lacks several large force-bearing categories present at the endpoints. This makes a configuration-balanced force loss different from a pooled-component loss. Leaving out each of the 11 intermediate categories in turn still selects $q^* = 0$, showing that this raw-reference outcome is not driven by one category. It does not validate that energy reference. With only one intermediate electronic temperature, exact endpoint anchoring and an interpolating curve also provide no empirical evidence at other intermediate temperatures.

Third, the diagnostic cells do not contain an isolated thermal spike. Periodic overlap, global heating, rapid thermostating, and the surface probe can dominate the measured volume. Once a selected region fills the cell, a volume-growth criterion can saturate even while temperatures become extreme. A finite-cell run that completes or has a nonpositive fitted volume slope is therefore not evidence of a stable bulk cascade. Small-cell defect counts likewise cannot establish preservation of the bulk damage-suppression effect reported by Wood et al. (2026). Increasing the number of directions in the same small cell would reduce sampling noise without removing these biases.

The numerical source comparison adds a fourth limitation: an exposure convention is needed to relate the plotted published values to the stated all-atom integral. The sigmoid has a nonzero cold floor, so its extensive exposure depends on the number of atoms even in the absence of a recoil. BECS removes that floor, and the small-cell pilot used to select its activation scale cannot by itself establish exposure matching in a much larger cell. This is another reason to keep offline matching, measured self-consistent matching, and a published exposure axis distinct. The source figures and their high-energy damage exception are retained explicitly in the numerical comparison.

The current evidence does not show a larger stable recoil range at matched self-consistent exposure, and does not establish lower Frenkel-pair counts and peak spike volume across the bulk recoil spectrum. Neither the failed calibration nor a small-cell heating comparison justifies those claims. A defensible larger study would first recover the exact energy-reference constants and file-level evaluation membership, specify the electronic-to-ionic temperature map, and verify spike containment with the published cell sizes. It would then compare all methods with the full direction ensemble, complete anneals, and measured exposure matching. These are experimental prerequisites, not results inferred from the present data.

The useful outcome is a separation of three questions that are easily conflated: whether an interpolation fits a stated electronic-energy reference, whether the implemented forces match that interpolation, and whether the resulting dynamics model dissipative irradiation physics. BECS supplies a precise and auditable answer to the first selection problem, while the present data expose its reference dependence. Direct LAMMPS checks establish the second. The velocity-reversal result and the finite-cell failures prevent inferring the third from smoothness or residency alone.

Reproducibility. The original switch form, criterion, split rule, blend grid, baseline thresholds, and intended recoil sweep were timestamped and hashed before the first potential evaluation. The

artifact retains that plan, subsequent deviations, and all failed execution attempts. Calibration stores per-configuration energies, endpoint forces, and quadratic sufficient statistics. Each MD trial retains its input, LAMMPS log, trajectory, and machine-readable outcome; completed trajectories have OVITO analysis outputs. Figures and tables are generated from these records. The runtime combines an official LAMMPS wheel with the unmodified supplied PACE library; no external scientific methodology code or new DFT calculation is used. Detailed code inspection and numerical provenance are provided in `codebase_notes.md`, `attempts_log.md`, and `logs/`.

References

- M. A. Wood, G. A. Koknat, and A. P. Thompson. Beyond classical molecular dynamics with layered interatomic potentials. Author-provided manuscript, supplemental information, and referee responses, 2026.
- Y. Lysogorskiy, C. van der Oord, A. Bochkarev, S. Menon, M. Rinaldi, T. Hammerschmidt, M. Mrovec, A. Thompson, G. Csányi, C. Ortner, and R. Drautz. Performant implementation of the atomic cluster expansion (PACE): Application to copper and silicon. arXiv:2103.00814, 2021. <https://arxiv.org/abs/2103.00814>.
- G. J. Ackland. Temperature dependence in interatomic potentials and an improved potential for Ti. *Journal of Physics: Conference Series*, 402:012001, 2012. doi:10.1088/1742-6596/402/1/012001.
- D. M. Duffy and A. M. Rutherford. Including the effects of electronic stopping and electron-ion interactions in radiation damage simulations. *Journal of Physics: Condensed Matter*, 19:016207, 2007. doi:10.1088/0953-8984/19/1/016207.
- E. Zarkadoula, D. M. Duffy, K. Nordlund, M. A. Seaton, I. T. Todorov, W. J. Weber, and K. Trachenko. Electronic effects in high-energy radiation damage in tungsten. *Journal of Physics: Condensed Matter*, 27:135401, 2015. doi:10.1088/0953-8984/27/13/135401.
- S. T. Murphy, S. L. Daraszewicz, Y. Giret, M. Watkins, A. L. Shluger, K. Tanimura, and D. M. Duffy. Dynamical simulations of an electronically induced solid-solid phase transformation in tungsten. *Physical Review B*, 92:134110, 2015. doi:10.1103/PhysRevB.92.134110.
- D. Immel, R. Drautz, and G. Sutmann. Adaptive-precision potentials for large-scale atomistic simulations. *Journal of Chemical Physics*, 162:114119, 2025. doi:10.1063/5.0245877.
- D. Immel, R. Drautz, and G. Sutmann. Conservative adaptive-precision interatomic potentials. arXiv:2512.07693, 2025. <https://arxiv.org/abs/2512.07693>.
- M.-C. Marinica, L. Ventelon, M. R. Gilbert, L. Proville, S. L. Dudarev, J. Marian, G. Bencteux, and F. Willaime. Interatomic potentials for modelling radiation defects and dislocations in tungsten. *Journal of Physics: Condensed Matter*, 25:395502, 2013. doi:10.1088/0953-8984/25/39/395502.
- LAMMPS developers. LAMMPS 22 July 2025 update 4: hybrid/scaled and ave/sphere/atom documentation and source. https://docs.lammps.org/pair_hybrid.html; https://docs.lammps.org/compute_ave_sphere_atom.html.
- OVITO developers. OVITO 3.16: Wigner-Seitz defect analysis and construct surface mesh. https://docs.ovito.org/reference/pipelines/modifiers/wigner_seitz_analysis.html; https://docs.ovito.org/python/modules/ovito_modifiers.html.